

Performance Driven Domain-level Data Acquisition

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Abstract—In many real-world domain adaptation problems, acquiring additional data is costly and constrained by practical considerations, such as purchasing data from external providers for model fine-tuning. Under such budget and information constraints, it is essential to identify which source domains are most informative for the target task. We propose PDA² (Performance Driven Adaptive Data Acquisition), a framework for domain-level data acquisition that adaptively selects which source domains to acquire data from in order to maximize performance improvement on a target task. We formulate the acquisition process as a multi-armed bandit problem and introduce an approximate utility estimator to assess each domain’s impact on the target loss. We derive a high-probability error bound for the utility approximation and show that PDA² achieves an $\mathcal{O}(\sqrt{N \log T/T})$ regret bound relative to the best source. Experiments demonstrate that PDA² identifies informative domains and outperforms baselines under budget constraints, yielding substantial gains in target performance. Code is available at <https://github.com/hanxinyan20/PDA2>.

Index Terms—multi-domain adaptation, data utility, model fine-tuning, data acquisition

I. INTRODUCTION

Models trained on a source distribution often suffer substantial performance degradation when deployed in a target domain due to domain shift. A standard approach to mitigating this issue is to acquire additional data for domain adaptation [1]–[3]. In practice, data are typically collected from multiple heterogeneous domains, such as different institutions, geographical regions, or third-party data vendors, leading to distinct data distributions [4]–[7]. In such settings, the effectiveness of data critically depends on which source domains are selected [8]–[10]. Source domains differ in their relevance to the target domain: some enable positive transfer, while others provide marginal benefit or induce negative transfer. Indiscriminate data acquisition can therefore degrade target performance or waste limited sampling budgets. These considerations motivate the study of multi-domain data acquisition under distribution shift, where data must be collected strategically to maximize adaptation effectiveness.

Existing studies related to multi-domain learning under distribution shift typically assume access to abundant data, or at least to a large pool of unlabeled samples. One prominent line studies data selection, where large pools of labeled (x, y) pairs are already available and the goal is to reduce training cost or improve efficiency by filtering examples according to

estimated contribution, influence, or representativeness [11]–[14]. Another line is active learning, which presumes abundant unlabeled inputs x and treats annotation as the dominant bottleneck; the core problem is to query labels for a subset of instances using acquisition criteria derived from predictive uncertainty or information-theoretic objectives [15]–[20].

In many practical settings, data acquisition differs from the two regimes discussed above. A representative example arises in model fine-tuning, where model developers often acquire domain-specific corpora to improve capabilities on downstream tasks, such as programming assistance or mathematical reasoning [21], [22]. Such data are rarely freely available at sufficient quality and scale: high-value corpora are often owned by third-party data vendors or content platforms, and must be accessed through paid licensing agreements. Before purchase, however, the learner usually cannot inspect the full candidate datasets and may only observe coarse descriptions, metadata, or small previews. Because providers differ in user populations, topics and annotation standards, their utility for the target task can vary substantially: some sources may yield strong downstream improvement, while others may be redundant, irrelevant, or even induce negative transfer. This setting is incompatible with active learning and data selection, which assume access to candidate instances before acquisition. It therefore gives rise to a domain-level data acquisition problem under budget constraints and uncertain utility, where the learner must adaptively allocate a limited budget to sources expected to bring the largest target-task improvement.

To address this challenge, we propose PDA² (Performance-Driven Adaptive Data Acquisition), a framework for domain-level data acquisition when no candidate data are accessible a priori. In contrast to existing methods that select samples from an available data pool, PDA² focuses on deciding where to acquire labeled data across heterogeneous source domains. We formulate this process as a Multi-Armed Bandit problem [23], where each data acquisition round involves estimating the utility of each source domain and sampling from the domain with the highest upper confidence bound on its utility. Inspired by gradient-based approaches [12], [24], we introduce an approximate utility that estimates each source domain’s impact on the target loss via a first-order approximation consistent with the optimizer’s update rule. This enables PDA² to adaptively prioritize the most informative domains under severe distribution shift and tight acquisition budgets. We summarize

our contributions as follows:

- 1) We introduce PDA², a performance-driven adaptive data acquisition framework that operates without requiring a centralized dataset (Section III), enabling deployment in real-world scenarios where data are limited and costly.
- 2) We offer a high-probability error bound for the utility approximation (Section IV-A) and prove that PDA² achieves an $\mathcal{O}(\sqrt{N \log T / T})$ regret bound with respect to the target domain performance (Section IV-B), ensuring convergence to the best-performing domain.
- 3) We empirically demonstrate that our approximate utility reliably distinguishes domain-level informativeness (Section V-A). We also show that PDA² outperforms baseline methods in terms of performance improvement in the target domain within a limited data budget (Section V-B–V-D), and remains effective with only a very small target validation set (Section V-E).

II. PROBLEM SETUP AND NOTATION

We denote a data distribution by P and a finite labeled dataset by $\mathcal{D}$. Let $\theta \in \mathbb{R}^d$ be the model parameter vector and $\ell(\theta; x, y)$ the per-sample loss. For any distribution P , the population risk is defined as

$$L_P(\theta) := \mathbb{E}_{(x,y) \sim P} [\ell(\theta; x, y)]. \quad (1)$$

Similarly, for any dataset $\mathcal{D}$, the empirical risk is defined as

$$L_{\mathcal{D}}(\theta) := \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \ell(\theta; x, y). \quad (2)$$

Given the parameters θ_{pre} of the pretrained model, we further define the average gradient of $\mathcal{D}$ evaluated at θ_{pre} as

$$g(\mathcal{D}) := \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \nabla_{\theta} \ell(\theta; x, y) \Big|_{\theta=\theta_{\text{pre}}}. \quad (3)$$

a) Domains and available feedback: We consider N source domains with distributions $P_1, \dots, P_N$ and a target domain with distribution P_* . To guide the acquisition process, we assume access to a small labeled validation set from the target domain,

$$\mathcal{D}_{\text{val}} = \{(x_j, y_j)\}_{j=1}^{n_{\text{val}}} \sim P_*^{n_{\text{val}}}, \quad (4)$$

which is used solely for evaluating target-domain performance during acquisition.

b) Domain-level acquisition protocol: The acquisition proceeds for T rounds. At each round $r \in \{1, \dots, T\}$, the learner selects a source domain $i_r \in \{1, \dots, N\}$ and acquires a labeled batch $\mathcal{S}_{i_r} \sim P_{i_r}^B$ of size B . After T rounds, the collected labeled training set is

$$\mathcal{D} = \bigcup_{r=1}^T \mathcal{S}_{i_r}, \quad (5)$$

which contains $|\mathcal{D}| = BT$ samples in total.

c) Fine-tuning: The learner fine-tunes the model θ_{pre} on $\mathcal{D}$ using a chosen optimizer. Let $\Delta_{\theta}(\mathcal{D})$ denote the resulting parameter update obtained by applying this optimizer to θ_{pre} on $\mathcal{D}$. The fine-tuned parameters are given by

$$\theta(\mathcal{D}) = \theta_{\text{pre}} + \Delta_{\theta}(\mathcal{D}). \quad (6)$$

III. METHOD

Our goal is to design a data acquisition policy that maximizes performance on the target domain under a fixed sampling budget BT . To this end, we first introduce a principled notion of utility for comparing different policies.

Fig. 1 provides an overview of PDA², which estimates performance-driven rewards from target validation feedback and uses a UCB policy to adaptively allocate the acquisition budget across source domains.

A. Acquisition policies and trajectories

An acquisition policy specifies a random sequence of source domains, referred to as a *trajectory*, denoted by $\pi = (i_1, \dots, i_T)$, where each index $i_r \in \{1, \dots, N\}$ indicates the source domain selected at step r . Conditional on a realized trajectory π , the training dataset is generated by drawing B samples from each selected domain. Formally, the resulting dataset satisfies

$$\mathcal{D}_{(\pi)} \sim \bigotimes_{r=1}^T P_{i_r}^B, \quad (7)$$

where $P_{i_r}^B$ denotes the distribution of B independent samples drawn from P_{i_r} . Thus, $\mathcal{D}_{(\pi)}$ is a random dataset of size BT whose distribution depends on the acquisition trajectory π .

B. Principled Utility Definition

Different trajectories produce different datasets, leading to different target-domain performance after model adaptation. We therefore quantify the utility of a trajectory by the reduction in target risk it induces.

Definition 1 (Utility of a trajectory). *We define the utility of a trajectory π as the expected reduction in target risk relative to the initial model parameters:*

$$\mathcal{U}(\pi) := L_{P_*}(\theta_{\text{pre}}) - \mathbb{E}_{\mathcal{D}_{(\pi)}} [L_{P_*}(\theta(\mathcal{D}_{(\pi)}))], \quad (8)$$

where the expectation is taken with respect to the randomness of the dataset $\mathcal{D}_{(\pi)}$ induced by the sampling process from domains specified by π .

Since a policy generates trajectories at random, its utility is measured by the expected trajectory utility.

Definition 2 (Utility of a policy). *The utility of an acquisition policy is defined by averaging over the randomness of trajectories induced by the policy:*

$$\mathcal{U} := \mathbb{E}_{\pi} [\mathcal{U}(\pi)]. \quad (9)$$

For comparison, we consider the special case where the trajectory always selects the same source domain i at every step, i.e., $\pi = (i, i, \dots, i)$. In this case, the corresponding dataset distribution is $\mathcal{D}_i \sim P_i^{BT}$. The utility of domain i is then obtained as the utility of this deterministic trajectory:

$$\mathcal{U}_i := L_{P_*}(\theta_{\text{pre}}) - \mathbb{E}_{\mathcal{D}_i} [L_{P_*}(\theta(\mathcal{D}_i))]. \quad (10)$$

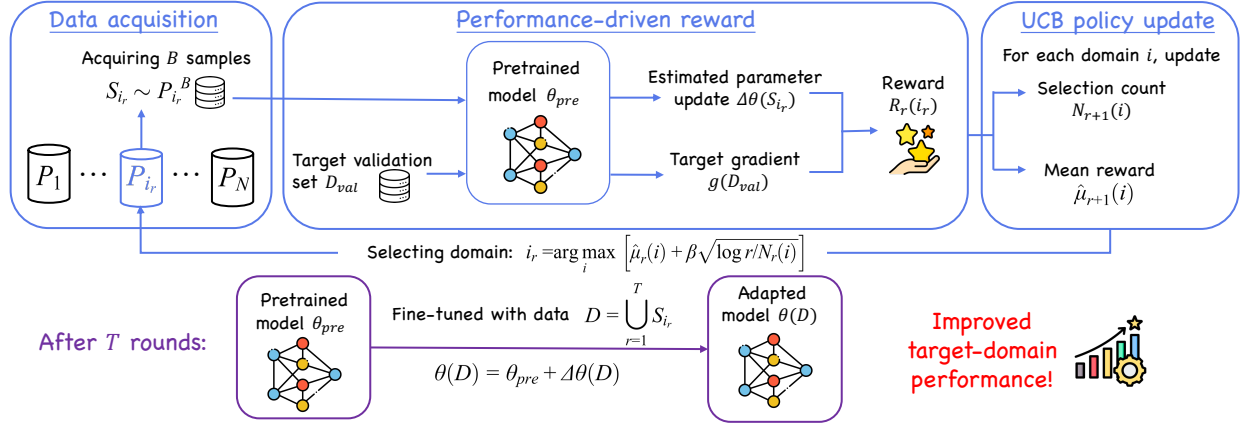


Fig. 1. Overview of PDA²: a performance-driven domain-level data acquisition framework that uses target validation feedback to estimate source-domain utility and adaptively selects informative domains via a UCB policy. The model is fine-tuned on all collected data after T acquisition rounds.

Definition 3 (Optimal domain). *We define the optimal domain as the domain with the highest utility:*

$$i_{\text{opt}} := \arg \max_{i \in \{1, \dots, N\}} \mathcal{U}_i, \quad (11)$$

with corresponding utility $\mathcal{U}_{i_{\text{opt}}}$.

The utilities $\mathcal{U}_i$ of individual source domains are not observable prior to data acquisition. If these values were available, the optimal strategy would be straightforward: allocate all samples to the domain i_{opt} that maximizes $\mathcal{U}_i$. Since this information is unavailable in practice, our objective is to design an acquisition policy whose expected utility $\mathcal{U}$ approaches $\mathcal{U}_{i_{\text{opt}}}$. In other words, the performance of the adaptive policy should, in expectation, converge to that of always sampling from the best single source domain.

C. Approximate Utility

Evaluating $\mathcal{U}(\pi)$ directly from Eq. (9) entails two major difficulties: (i) it requires repeatedly retraining the model on many realizations of $\mathcal{D}(\pi)$, which is computationally prohibitive; and (ii) it involves estimating the expected risk $L_{P_\star}$ under the unknown target distribution $P_\star$. To obtain a feasible estimator, we adopt two approximations. First, we approximate the intractable expectation over $P_\star$ using the validation set $\mathcal{D}_{\text{val}}$ of n_{val} i.i.d. samples from $P_\star$. Second, we approximate the change in validation loss by applying a first-order Taylor expansion of $L_{\mathcal{D}_{\text{val}}}(\theta)$ around the pre-trained parameters θ_{pre} : $L_{\mathcal{D}_{\text{val}}}(\theta_{\text{pre}} + \Delta\theta(\mathcal{D}(\pi))) \approx L_{\mathcal{D}_{\text{val}}}(\theta_{\text{pre}}) + g(\mathcal{D}_{\text{val}})^\top \Delta\theta(\mathcal{D}(\pi))$.

Definition 4 (Approximate utility of a trajectory). *Given a validation set $\mathcal{D}_{\text{val}}$, which consists of n_{val} i.i.d. samples from the target domain $P_\star$, the approximate utility of trajectory π is*

$$\tilde{\mathcal{U}}(\pi) := -\mathbb{E}_{\mathcal{D}(\pi)} [g(\mathcal{D}_{\text{val}})^\top \Delta\theta(\mathcal{D}(\pi))], \quad (12)$$

where the expectation is over datasets $\mathcal{D}(\pi)$ generated under trajectory π .

This approximate utility quantifies the expected decrease in validation loss induced by an acquisition trajectory π . It

requires only gradient evaluations, avoiding repeated retraining. Its approximation error arises from two sources: (i) the generalization error between the finite validation set $\mathcal{D}_{\text{val}}$ and the true target distribution $P_\star$, and (ii) the truncation error from neglecting the second-order term in the Taylor expansion. We establish a high-probability bound on the approximation error in Section IV-A.

To make the update operator $\Delta\theta(\mathcal{D})$ concrete, we illustrate it using the widely adopted Adam optimizer. In our setting, parameters are updated once every T acquisition rounds: after collecting B samples per round, the accumulated batch of BT samples is used to update θ_{t-1} to θ_t . Here, t indexes the number of parameter updates rather than acquisition rounds. Given stored moment estimates $\mathbf{m}_{t-1}, \mathbf{v}_{t-1}$ at θ_{pre} , the bias-corrected moments are

$$\begin{aligned} \hat{\mathbf{m}}_t &= \frac{\beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) g(\mathcal{D})}{1 - \beta_1^t}, \\ \hat{\mathbf{v}}_t &= \frac{\beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) g(\mathcal{D}) \odot g(\mathcal{D})}{1 - \beta_2^t}. \end{aligned} \quad (13)$$

The Adam step is then

$$\Delta\theta(\mathcal{D}) = -\alpha \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon}. \quad (14)$$

where α is the learning rate, β_1, β_2 are the EMA coefficients, $\epsilon > 0$ is a small constant, and $\odot$ denotes elementwise product. In the case of stochastic gradient descent (SGD), the update takes the simpler form

$$\Delta\theta(\mathcal{D}) = -\alpha g(\mathcal{D}). \quad (15)$$

Although our subsequent analysis is presented under the Adam update rule Eq. (14), the theoretical bounds can be easily transferred to the SGD setting Eq. (15).

D. Acquisition as a Multi-Armed Bandit Problem

The approximate utility introduced in the previous section quantifies how informative each source domain is for the target task. However, these utilities are unknown a priori, before any data are collected. Consequently, our goal is to sequentially

decide, at each acquisition round, which domain to sample from so as to maximize the expected utility over all rounds.

a) *Bandit formulation*: This sequential decision-making problem can be naturally framed as a multi-armed bandit problem. In this bandit framework, each source domain corresponds to an arm, and selecting a domain yields a stochastic reward reflecting the informativeness of the sampled data to the target domain. The objective is to acquire samples across domains to maximize the cumulative reward. A central challenge is the exploration–exploitation trade-off: the algorithm must explore uncertain domains to estimate their utility, while exploiting domains that appear promising based on past observations.

b) *Feedback from each domain*: To apply this bandit framework, we need a per-round feedback signal indicating how helpful the sampled batch is for the target task. Recall that the approximate utility (Eq. (12)) depends on the alignment between the validation gradient and the update induced by new data. This motivates the following reward definition:

Definition 5 (Per-round reward). *At acquisition round r , when the algorithm selects domain $i_r \in \{1, \dots, N\}$ and draws a batch $S_{i_r} \sim P_{i_r}^B$, the reward of this action is defined as*

$$R_r(i_r) = -g(\mathcal{D}_{\text{val}})^\top \Delta_\theta(S_{i_r}), \quad (16)$$

where $g(\mathcal{D}_{\text{val}})$ is the gradient on the target validation set, and $\Delta_\theta(S_{i_r})$ is the parameter update produced by the optimizer.

A larger reward indicates that the batch is expected to reduce the validation loss more effectively. $R_r(i_r)$ serves as the stochastic payoff for selecting domain i_r .

c) *Domain selection via UCB*: We adopt the Upper Confidence Bound (UCB) strategy to balance exploration and exploitation. Let $N_r(i)$ be the number of times domain i has been sampled before round r , and let $\hat{\mu}_r(i)$ denote its empirical mean reward:

$$N_r(i) = \sum_{j=1}^{r-1} \mathbf{1}\{i_j = i\}, \quad (17)$$

$$\hat{\mu}_r(i) = \frac{1}{N_r(i)} \sum_{j=1}^{r-1} \mathbf{1}\{i_j = i\} R_j(i).$$

Each domain is sampled once in the first N rounds. For $r > N$, we select

$$i_r \in \arg \max_{i \in \{1, \dots, N\}} \left\{ \hat{\mu}_r(i) + \beta \sqrt{\frac{\log r}{N_r(i)}} \right\}, \quad (18)$$

where $\beta > 0$ controls the degree of exploration. After observing $R_r(i_r)$, the statistics are updated as

$$N_{r+1}(i_r) \leftarrow N_r(i_r) + 1, \quad (19)$$

$$\hat{\mu}_{r+1}(i_r) \leftarrow \hat{\mu}_r(i_r) + \frac{R_r(i_r) - \hat{\mu}_r(i_r)}{N_{r+1}(i_r)}.$$

d) *Interpretation*: In early rounds PDA² explores domains with little data (high uncertainty), while in later rounds it concentrates on domains that yield high rewards. This policy gradually focuses on the most informative domain, approaching the performance of the best fixed-domain strategy.

IV. THEORETICAL ANALYSIS

In this section, we first establish a bound on the approximation error between the approximate utility $\tilde{\mathcal{U}}(\pi)$ and the true utility $\mathcal{U}(\pi)$, as formalized in Theorem 1. We then provide the regret of the adaptive acquisition policy over T acquisition rounds in Theorem 2, demonstrating that the policy's performance converges to that of the best single domain.

Our analysis relies on the following assumptions, which hold for a wide range of loss functions and models.

Assumption 1 (Smoothness). *There exists a constant $L_1 > 0$ such that for all samples (x, y) and parameters θ, θ' , we have*

$$\|\nabla_\theta \ell(\theta; x, y) - \nabla_\theta \ell(\theta'; x, y)\|_2 \leq L_1 \|\theta - \theta'\|_2.$$

Assumption 2 (Bounded element-wise gradients). *There exists a constant $G > 0$ such that for all samples (x, y) and parameters θ , we have*

$$\|\nabla_\theta \ell(\theta; x, y)\|_\infty \leq G.$$

A. Bounding the error of utility approximation

We now provide a high-probability bound on the discrepancy between the true utility $\mathcal{U}(\pi)$ and its approximation $\tilde{\mathcal{U}}(\pi)$.

Theorem 1 (Approximation error). *Let d be the dimensionality of the parameter vector, α the learning rate, and n_{val} the size of the validation set drawn i.i.d. from the target distribution P_* . Assume Assumption 1 and 2 hold, and the Adam hyperparameters satisfy $\beta_1 < \beta_2 < 1$. Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the randomness in the validation set, it holds that*

$$|\tilde{\mathcal{U}}(\pi) - \mathcal{U}(\pi)| \leq (\alpha C_1 + \alpha^2 C_2) \sqrt{\frac{\log(2/\delta)}{2n_{\text{val}}}} + \frac{1}{2} \alpha^2 C_2, \quad (20)$$

where $C_1 = 2dG\sqrt{\frac{1-\beta_1}{1-\beta_2}}$ and $C_2 = L_1 d \cdot \frac{1-\beta_1}{1-\beta_2}$.

Theorem 1 quantifies the difference between the true utility $\mathcal{U}(\pi)$ and its approximation $\tilde{\mathcal{U}}(\pi)$. There are two sources of approximation error, the first is the high-probability generalization error $(\alpha C_1 + \alpha^2 C_2) \sqrt{\frac{\log(2/\delta)}{2n_{\text{val}}}}$, which captures the error incurred by using a finite validation set instead of the target population. The second is the truncation error $\frac{1}{2} \alpha^2 C_2$, resulting from ignoring higher-order terms in the Taylor expansion of the loss. These two sources provide an upper bound on the total approximation error of $\tilde{\mathcal{U}}(\pi)$ with high probability. The detailed proof is provided in Appendix A-A.

B. Bounding the utility regret

We now analyze the performance of our adaptive acquisition policy by comparing its utility against that of the best single-domain strategy, i.e. $\mathcal{U}$ versus $\mathcal{U}_{\text{opt}}$.

Theorem 2 (Utility regret bound). *Let N be the number of source domains and T the number of acquisition rounds. Suppose Assumptions 1 and 2 hold, and the Adam hyperparameters satisfy $\beta_1 < \beta_2 < 1$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random draw of the*

validation set $\mathcal{D}_{\text{val}}$, the expected utility regret of the proposed data acquisition algorithm is bounded by:

$$\mathcal{U}_{i_{\text{opt}}} - \mathcal{U} \leq \mathcal{O}\left(\sqrt{\frac{N \log T}{T}}\right)$$

provided that the learning rate α satisfies:

$$\alpha \leq \mathcal{O}\left(\min\left\{\left(\frac{N \log T}{T}\right)^{1/4}, \sqrt{\frac{N \cdot n_{\text{val}} \cdot \log T}{T(T \log N + \log(4/\delta))}}\right\}\right).$$

Theorem 2 shows that the utility of our algorithm converges to that of the best single-domain strategy. The convergence rate $\mathcal{O}\left(\sqrt{\frac{N \log T}{T}}\right)$ matches the standard regret bounds in stochastic multi-armed bandits. Intuitively, although the algorithm does not know the best domain in advance, this regret bound guarantees it gradually learns to concentrate on high-utility domains, and its performance approaches that of always sampling from i_{opt} . Note that this regret bound is not a straightforward consequence of classical UCB analysis. Two technical obstacles arise in our setting: (i) the Adam update (Eq. (14)) involves the batch gradient $g(\mathcal{D})$ in its denominator, so the per-round reward $R_r(i)$ —computed on B samples—does not have an expectation equal to the approximate utility $\tilde{\mathcal{U}}_i$, which is defined using BT samples; (ii) our result bounds the regret in terms of the *true utility* $\mathcal{U}$ rather than the approximate utility $\tilde{\mathcal{U}}$. Moreover, the domain that maximizes $\tilde{\mathcal{U}}_i$ need not coincide with the domain that maximizes $\mathcal{U}_i$. Overcoming these gaps requires novel analysis to bridge batch-level rewards and trajectory-level true utility, which is a key contribution of our theoretical results. A complete proof is provided in Appendix A-B.

V. EXPERIMENTS

We now evaluate our adaptive acquisition algorithm through real-world experiments. Recall that the reward $R_r(i_r)$ depends on the parameter update $\Delta_{\theta}(\cdot)$, which is determined by the optimizer. Accordingly, we implement two variants of PDA²:

- PDA²(Adam): Uses the Adam optimizer (Eq. (14)) in the reward calculation.
- PDA²(SGD): Replaces Adam with the SGD optimizer (Eq. (15)) to compute the reward.

We compare our method against various domain-level data acquisition baselines:

- Uniform: Samples source domains uniformly.
- Quad [11]: Balances influence and diversity to enhance training efficiency.
- DAA [25]: Reweights source domains adaptively to minimize the MMD with respect to the target domain.

Additionally, we compare with classic active learning methods:

- Margin, Entropy [26], and LeastConfidence [15]: Use uncertainty sampling to select informative instances.
- BADGE [16]: Combines uncertainty and diversity via gradient-based clustering.

Our setting differs from classical active learning, which assumes access to a large unlabeled candidate pool and selects

TABLE I
KENDALL’S τ CORRELATION BETWEEN ESTIMATED APPROXIMATE UTILITY AND ESTIMATED TRUE UTILITY ON VALIDATION AND TEST SPLITS. EACH SETTING IS REPEATED 10 TIMES, AND WE REPORT THE MEAN AND STANDARD ERROR.

Method	Val	Test
FL→NY		
Adam-based (ours)	0.91 ± 0.05	0.91 ± 0.08
SGD-based (ours)	0.84 ± 0.09	0.85 ± 0.09
Influence (baseline)	0.64 ± 0.12	0.64 ± 0.10
NC→TX		
Adam-based (ours)	0.93 ± 0.05	0.91 ± 0.06
SGD-based (ours)	0.87 ± 0.08	0.86 ± 0.08
Influence (baseline)	0.04 ± 0.29	0.02 ± 0.28

individual instances for annotation. In contrast, PDA² selects source domains before observing the candidate samples. We nevertheless include active learning methods for broader comparison and evaluate them under a more favorable setting. All experiments in this section are repeated over 10 independent runs, and we report the mean and standard deviation.

A. Approximate Utility Reliably Reflects True Domain Utility

We test whether the proposed approximate utility can correctly identify which source domains are most beneficial for transfer. We compare approximate utility scores computed from Adam and SGD update rules with the estimated true utility of each source domain. We also include an influence-function score as an external baseline.

a) *Setup*: We use the US Accident dataset [27], which records traffic accidents across U.S. states. The task is to predict whether an accident is severe based on features such as weather conditions and road type. Each transfer setting specifies one state for pre-training, one state as the target state and other eight states as candidate source domains. The model is first trained on the full training set in the pre-training state to obtain θ_{pre} . For each candidate source domain i , we estimate its approximate utility using a batch of $BT = 256$ i.i.d. samples $\mathcal{D}_i$ drawn from this domain. Formally, we compute:

$$\tilde{\mathcal{U}}_i(\mathcal{D}_i) = -g(\mathcal{D}_{\text{val}})^\top \Delta_{\theta}(\mathcal{D}_i), \quad (21)$$

where Δ_{θ} is the parameter update, computed via the Adam rule (Eq. (14)) or the SGD rule (Eq. (15)), resulting in two variants. The expected value over the batch is exactly the approximate utility, $\mathbb{E}[\tilde{\mathcal{U}}(\mathcal{D}_i)] = \tilde{\mathcal{U}}_i$.

As an external baseline, we also compute an influence-function score by averaging per-sample influence [28].

$$\tilde{\mathcal{U}}_i(\mathcal{D}_i) = \frac{1}{|\mathcal{D}_i|} \sum_{(x,y) \in \mathcal{D}_i} -g(\mathcal{D}_{\text{val}})^\top H_{\theta_{\text{pre}}}^{-1} g(x,y). \quad (22)$$

Here, $H_{\theta_{\text{pre}}}$ denotes the Hessian of the training loss with respect to model parameters at θ_{pre} .

The estimated true utilities are obtained by fine-tuning the pre-trained model on $\mathcal{D}_i$ and measuring the reduction in target risk on validation and test splits.

b) *Results*: We rank source domains by (i) their estimated approximate utility scores $\tilde{\mathcal{U}}_i(\mathcal{D}_i)$ (Eq. (21) or Eq. (22)) and (ii) their estimated true utilities. We report the Kendall rank correlation coefficient τ between the two rankings in

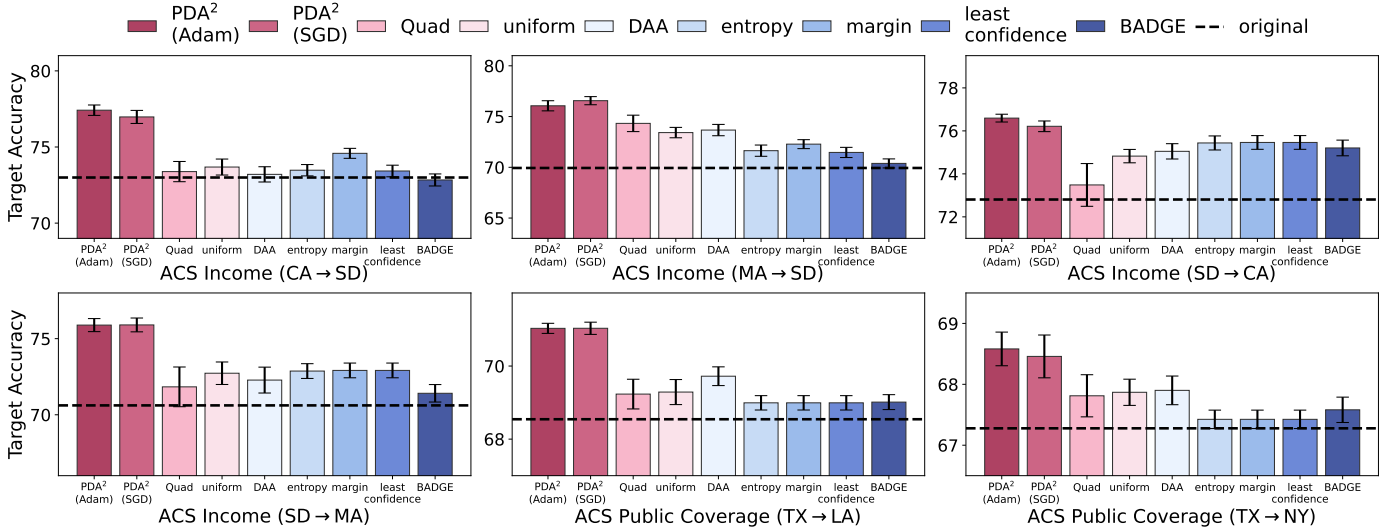


Fig. 2. Performance of different methods on ACS Income and ACS Public Coverage. The black dashed line is test accuracy of the pretrained model.

Table I. All fine-tuning experiments are conducted with the Adam optimizer. The Adam-based variant achieves the highest rank correlations ($\tau \approx 0.9$), outperforming the SGD-based variant. In contrast, the influence-function baseline performs poorly, with unstable correlations. *These results underscore three findings: (i) our approximate utility provides a reliable signal of domain usefulness, (ii) correlation improves when the approximate utility is instantiated with the same optimizer used in model fine-tuning, and (iii) validation performance strongly aligns test performance, supporting its use as a proxy for guiding acquisition.*

B. Case Study 1: Socioeconomic Prediction

a) *Setup*: We use ACS Income and ACS Public Coverage to evaluate domain-wise data acquisition for regional adaptation. Each U.S. state is treated as one domain. For each transfer setting, we designate one state as the pre-training domain, one state as the target domain, and $N = 8$ other states as candidate source domains. We train the initial model on the full set of the pre-training state to obtain θ_{pre} . From the target state, 50 labeled samples are used as the validation set for guiding acquisition, and the remaining samples are held out for testing. At each acquisition round, the algorithm first chooses one candidate source state. Only after this choice is made, $B = 4$ labeled samples are collected from the chosen state. Therefore, the decision is made at the domain level: the algorithm decides which state to sample from, rather than selecting individual examples from an already available labeled pool. We run $T = 32$ rounds, yielding 128 acquired source samples in total. The model is fine-tuned on these acquired samples, and target accuracy is evaluated on the held-out target test set.

b) *Results*: We show all six transfer settings and the sampling ratios across source domains in Table II. In the ACS Public Coverage task (TX → NY), PDA² outperforms baselines by selectively acquiring from CA, which shares policy-relevant patterns with NY. Similarly, in ACS Income (CA → SD), PDA²

TABLE II
SAMPLING RATIO ACROSS MULTIPLE SOURCE DOMAINS OF PDA²(ADAM) ON SIX TRANSFER SETTINGS IN CASE STUDY 1.

Task.Shift	1	2	3	4	5	6	7	8
Inc.CA → SD	CT.040	AR. 434	MD.044	MA.050	ID.172	HI.116	NY.069	WA.075
Inc.MA → SD	CT.051	AR. 295	MD.049	CA.074	ID.183	HI.089	NY.100	WA.160
Inc.SD → CA	PR.036	KS.110	MA. 309	ND.150	NY.165	ME.088	WY.060	WV.083
Inc.SD → MA	PR.034	KS.077	CA.202	ND.148	NY. 254	ME.060	WY.123	WV.102
Cov.TX → NY	UT.069	GA.046	KS.122	CA. 404	NE.072	SD.134	ID.075	WY.077
Cov.TX → LA	UT.040	GA.109	KS.106	CA. 440	NE.069	SD.098	ID.074	WY.065

TABLE III
AVERAGE RUNTIME OVERHEAD PER SAMPLE (IN MILLISECONDS) FOR VARIOUS DATA ACQUISITION METHODS ON THE ACS INCOME (CA → SD) TASK. POOL-SIZE DEPENDENT INDICATES WHETHER THE METHOD'S COMPUTATION TIME SCALES WITH THE UNLABELED POOL SIZE.

Method	Overhead (ms/sample)	Pool-size Dependent?
PDA ² (Adam)	1.088	No
PDA ² (SGD)	0.760	No
uniform	0.0217	No
entropy	5.433	Yes
margin	6.000	Yes
least confidence	5.240	Yes
Quad	12.105	No
BADGE	34.450	Yes
DAA	145.523	No

favors AR, reflecting similarities to target state SD in income-prediction structures. The acquisition behaviors align with domain-level socioeconomic similarities, offering an interpretable rationale for why transfer succeeds. We report the overhead incurred by different data acquisition approaches on the ACS Income (CA → SD) task in Table III. As shown in Figure 2, PDA² achieves the highest or near-highest target accuracy across all six settings. *PDA² reliably identifies and acquires from the most beneficial source domains under distribution shifts, enabling effective adaptation.*

C. Case Study 2: Evolving Traffic Prediction

a) *Setup*: We use the US Accident dataset to evaluate adaptive acquisition in evolving traffic prediction. Each state

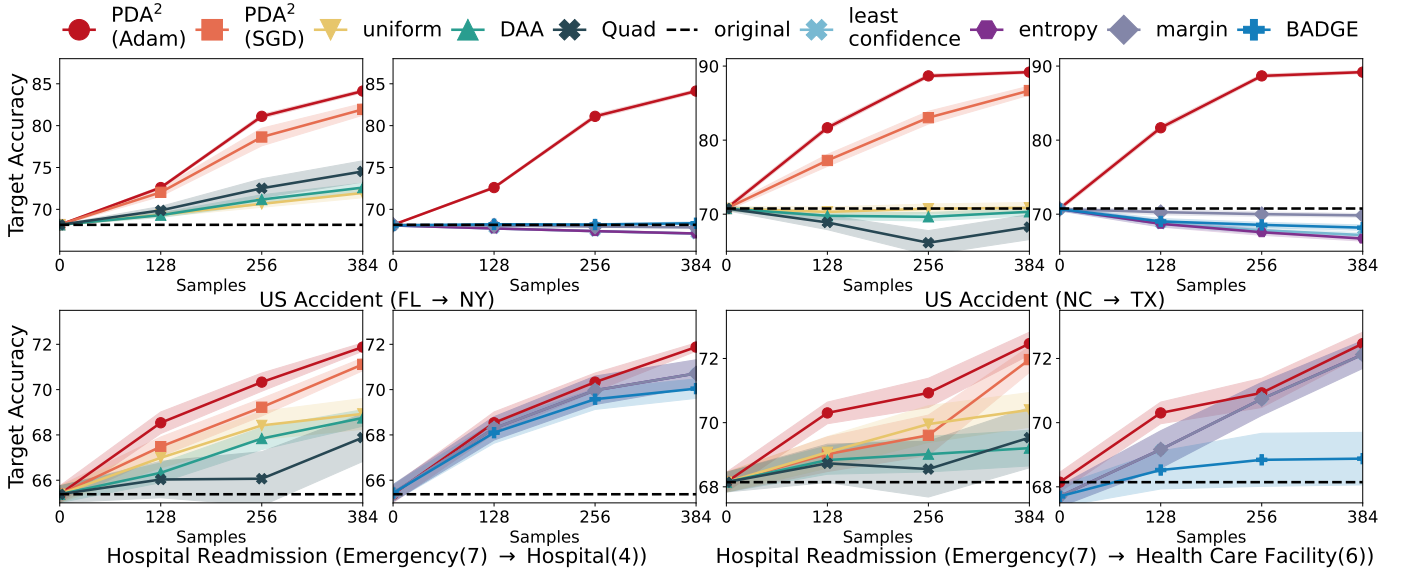


Fig. 3. Performance of different methods for multi-stage data acquisition on tabular prediction task.

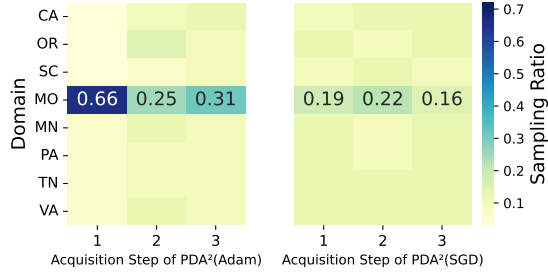


Fig. 4. Sampling ratios of PDA^2 (Adam) and PDA^2 (SGD) over acquisition steps on US Accident(NC→TX) shifts.

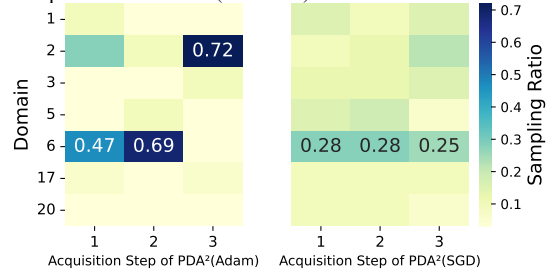


Fig. 5. Sampling ratios of PDA^2 (Adam) and PDA^2 (SGD) over acquisition steps on Hospital Readmission (7→4) shifts.

is treated as one domain. We consider two transfer settings: Accident (FL→NY), where the model is pretrained on Florida (FL), adapted to New York (NY), and acquires data from CA, OR, TX, SC, MN, PA, TN, and VA; and Accident (NC→TX), where the model is pretrained on North Carolina (NC), adapted to Texas (TX), and acquires data from CA, OR, SC, MO, MN, PA, TN, and VA. As the model is updated, the marginal utility of each source state may change because previously useful domains can become saturated, while other domains may become more informative for the current model. We therefore divide acquisition into $S = 3$ stages and restart the acquisition policy at the beginning of each stage. Within each stage, the

algorithm runs for $T = 32$ rounds, selecting one source state per round and acquiring $B = 4$ labeled samples. After each stage, the model is fine-tuned using the data acquired in that stage, and the next stage re-estimates source-domain utilities for the updated model.

b) Results: Fig. 3 shows that PDA^2 (Adam) improves more rapidly than the baselines. Fig. 4 further illustrates that (i) PDA^2 (Adam) identifies informative domains with fewer samples, and (ii) the sampling ratios gradually converge to uniformity, reflecting utility saturation and aligning with the diminishing accuracy gains observed in Fig. 3. PDA^2 (Adam) mitigates sampling redundancy by identifying when further samples from a domain offer diminishing utility.

D. Case Study 3: Hospital Readmission Prediction

a) Setup: We use the Hospital Readmission dataset to evaluate multi-stage acquisition in healthcare adaptation. The task is to predict whether a patient will be readmitted within 30 days of discharge. We treat each admission source as one domain, since patients entering from different sources, such as emergency rooms, physician referrals, or hospital transfers, may exhibit different clinical profiles and readmission risks. We consider two transfer settings. In Hospital Readmission (7→4), the model is pretrained on Emergency Room (7), adapted to Transfer from a hospital (4), and acquires data from Physician Referral (1), Clinic Referral (2), HMO Referral (3), Transfer from a Skilled Nursing Facility (5), Transfer from another health care facility (6), NULL (17), and Not Mapped (20). In Hospital Readmission (7→6), the model is again pretrained on Emergency Room (7), adapted to Transfer from another health care facility (6), and acquires data from Physician Referral (1), Clinic Referral (2), HMO Referral (3), Transfer from a hospital (4), Transfer from a Skilled Nursing Facility (5), NULL (17), and Not Mapped (20). We follow the same multi-stage protocol as in Case Study 2.

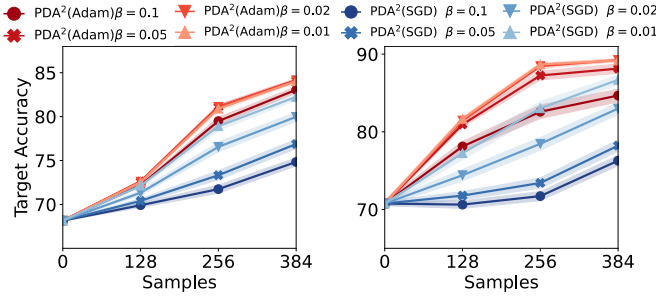


Fig. 6. Impact of varying β on the Accident (FL $\rightarrow$ NY) and (NC $\rightarrow$ TX) shifts.

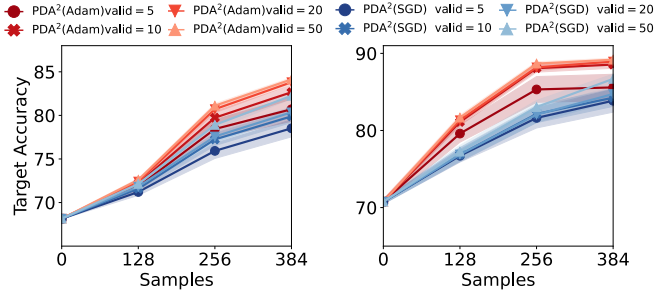


Fig. 7. Impact of varying validation size on the Accident (FL $\rightarrow$ NY) and (NC $\rightarrow$ TX) shifts.

b) *Results:* As shown in Fig. 3 (row 2), $PDA^2(\text{Adam})$ shows advantages compared with other baselines especially in early stage. It initially prioritizes Health Care Facility (6), but shifts toward Clinic Referral (2) once the former's utility diminishes (see Fig. 5). This illustrates PDA^2 's ability to track changing marginal utilities as the model evolves. $PDA^2(\text{Adam})$ adapts to evolving domain utilities over time, enabling effective use of data budgets in dynamic settings.

E. Ablation Studies

a) *Sensitivity to β :* We first examine the sensitivity of PDA^2 to the trade-off parameter β in Eq. (19), which controls the balance between exploring uncertain domains and exploiting domains with high estimated immediate utility. A practical acquisition strategy should not rely on a narrowly tuned value of β . We vary $\beta \in \{0.01, 0.02, 0.05, 0.1\}$ on the two settings in Section V-C. As shown in Fig. 6, $PDA^2(\text{Adam})$ is highly robust to this parameter. Moderate perturbations of β do not substantially affect performance. Performance drops noticeably only when β becomes very large, e.g., $\beta = 0.1$. Even under this unfavorable setting, $PDA^2(\text{Adam})$ remains competitive and matches the best performance of $PDA^2(\text{SGD})$. These results suggest that the Adam-based update provides a more stable and effective optimization signal, making PDA^2 less sensitive to the exact choice of β .

b) *Robustness to limited validation data:* We next evaluate $PDA^2(\text{Adam})$ under extremely small validation sets. In many practical adaptation scenarios, only a few labeled validation samples are available for estimating domain utility. Such scarcity makes the utility estimates noisy and may destabilize the acquisition policy. As shown in Fig. 7, $PDA^2(\text{Adam})$ remains robust even when the validation set contains only

5 or 10 samples. This indicates that $PDA^2(\text{Adam})$ can still obtain informative adaptation signals from scarce validation feedback, rather than relying on a large validation set.

VI. DISCUSSION

In this paper, we introduce PDA^2 , a performance-driven adaptive data acquisition framework for multi-source domain adaptation. By formulating the acquisition process as a multi-armed bandit problem and introducing a tractable approximation of utility, PDA^2 acquires the most informative samples without relying on a pre-collected data pool, making it particularly suitable for data-scarce settings. The effectiveness of PDA^2 relies on the assumption that at least one source domain is sufficiently informative with respect to the target domain. An important direction for future work is to relax this assumption by developing strategies that integrate complementary knowledge from diverse source domains.

APPENDIX A PROOF

A. Proof of Total Approximation Error

Lemma 1 (Adam step-size bound). *For the Adam update at step t , assume $0 < \beta_1 \leq \beta_2 < 1$ and $\epsilon > 0$. Then, for every coordinate i and dataset $\mathcal{D}$,*

$$|\Delta_{\theta_i}(\mathcal{D})| \leq \alpha C, \quad C = \sqrt{\frac{1 - \beta_1}{1 - \beta_2}}.$$

Proof. Let $p_k = \frac{(1 - \beta_1)\beta_1^{t-k}}{1 - \beta_1^t}$ and $q_k = \frac{(1 - \beta_2)\beta_2^{t-k}}{1 - \beta_2^t}$. Then we have

$$\hat{m}_{t,i} = \sum_{k=1}^t p_k g_{k,i}, \quad \hat{v}_{t,i} = \sum_{k=1}^t q_k g_{k,i}^2.$$

If $\hat{v}_{t,i} = 0$, then $g_{k,i} = 0$ for all k , and hence $\hat{m}_{t,i} = 0$, so the update is zero. Otherwise,

$$|\Delta_{\theta_i}| = \alpha \frac{|\hat{m}_{t,i}|}{\sqrt{\hat{v}_{t,i}} + \epsilon} \leq \alpha \frac{|\hat{m}_{t,i}|}{\sqrt{\hat{v}_{t,i}}}.$$

By weighted Cauchy-Schwarz,

$$\frac{\hat{m}_{t,i}^2}{\hat{v}_{t,i}} \leq \sum_{k=1}^t \frac{p_k^2}{q_k} = \frac{(1 - \beta_1)^2(1 - \beta_2^t)}{(1 - \beta_1^t)^2(1 - \beta_2)} \sum_{j=0}^{t-1} \left(\frac{\beta_1^2}{\beta_2}\right)^j \leq \frac{1 - \beta_1}{1 - \beta_2}.$$

Taking square roots proves the claim. $\square$

Definition 6 (Validation-set risk reduction for a trajectory). *Given a validation set $\mathcal{D}_{\text{val}}$ and a trajectory π , define*

$$\mathcal{U}_{\text{val}}(\pi) = L_{\mathcal{D}_{\text{val}}}(\theta_{\text{pre}}) - \mathbb{E}_{\mathcal{D}(\pi)} [L_{\mathcal{D}_{\text{val}}}(\theta(\mathcal{D}(\pi)))],$$

where the expectation is over the randomness of the training dataset $\mathcal{D}(\pi)$ generated by π .

Lemma 2 (Truncation error). *For any validation set $\mathcal{D}_{\text{val}}$ and trajectory π , under Assumption 1 and Lemma 1,*

$$|\mathcal{U}_{\text{val}}(\pi) - \tilde{\mathcal{U}}(\pi)| \leq \frac{L_1}{2} d\alpha^2 C^2.$$

Proof. Let $\Delta_\pi = \theta(\mathcal{D}(\pi)) - \theta_{\text{pre}}$. A second-order Taylor expansion of $L_{\mathcal{D}_{\text{val}}}(\cdot)$ at θ_{pre} gives

$$L_{\mathcal{D}_{\text{val}}}(\theta(\mathcal{D}(\pi))) = L_{\mathcal{D}_{\text{val}}}(\theta_{\text{pre}}) + g(\mathcal{D}_{\text{val}})^\top \Delta_\pi + \frac{1}{2} \Delta_\pi^\top \nabla^2 L_{\mathcal{D}_{\text{val}}}(\xi_\pi) \Delta_\pi.$$

for some point ξ_π on the line segment between θ_{pre} and $\theta(\mathcal{D}(\pi))$. Substituting this expansion into the definition of $\mathcal{U}_{\text{val}}(\pi)$ and comparing it with $\tilde{\mathcal{U}}(\pi)$, the difference is exactly the expected second-order remainder. Therefore,

$$|\mathcal{U}_{\text{val}}(\pi) - \tilde{\mathcal{U}}(\pi)| \leq \frac{L_1}{2} \mathbb{E}_{\mathcal{D}(\pi)} \|\Delta_\pi\|_2^2.$$

By Lemma 1, $\|\Delta_\pi\|_2^2 \leq d\alpha^2 C^2$, which proves the result. $\square$

Proof of Theorem 1. We decompose the total approximation error as

$$|\tilde{\mathcal{U}}(\pi) - \mathcal{U}(\pi)| \leq \underbrace{|\mathcal{U}(\pi) - \mathcal{U}_{\text{val}}(\pi)|}_{\text{(a) generalization error}} + \underbrace{|\mathcal{U}_{\text{val}}(\pi) - \tilde{\mathcal{U}}(\pi)|}_{\text{(b) truncation error}}.$$

Term (a) comes from replacing the population distribution $P_\star$ by the finite validation set, while term (b) comes from using a first-order Taylor approximation of the validation utility.

We first bound term (a). Let $\Delta_\pi = \theta(\mathcal{D}(\pi)) - \theta_{\text{pre}}$. For any sample (x, y) , Taylor's expansion and Assumptions 1–2 imply

$$|\ell(\theta(\mathcal{D}(\pi)); x, y) - \ell(\theta_{\text{pre}}; x, y)| \leq G \|\Delta_\pi\|_1 + \frac{L_1}{2} \|\Delta_\pi\|_2^2.$$

By Lemma 1, we have $\|\Delta_\pi\|_1 \leq d\alpha C$ and $\|\Delta_\pi\|_2^2 \leq d\alpha^2 C^2$. Hence the absolute loss change is bounded by

$$B = dG\alpha C + \frac{L_1}{2} d\alpha^2 C^2.$$

Define $\phi_\pi(x, y) = \mathbb{E}_{\mathcal{D}(\pi)} [\ell(\theta_{\text{pre}}; x, y) - \ell(\theta(\mathcal{D}(\pi)); x, y)]$. Then $|\phi_\pi(x, y)| \leq B$, and

$$\mathcal{U}(\pi) = \mathbb{E}_{(x, y) \sim P_\star} [\phi_\pi(x, y)], \quad \mathcal{U}_{\text{val}}(\pi) = \frac{1}{n_{\text{val}}} \sum_{j=1}^{n_{\text{val}}} \phi_\pi(x_j, y_j).$$

Since the validation samples are i.i.d., Hoeffding's inequality gives, with probability at least $1 - \delta$,

$$|\mathcal{U}(\pi) - \mathcal{U}_{\text{val}}(\pi)| \leq 2B \sqrt{\frac{\log(2/\delta)}{2n_{\text{val}}}}.$$

Term (b) is bounded by $|\mathcal{U}_{\text{val}}(\pi) - \tilde{\mathcal{U}}(\pi)| \leq \frac{L_1}{2} d\alpha^2 C^2$ (Lemma 2).

Adding the generalization error (a) and the truncation error (b) yields the total approximation error bound. $\square$

B. Proof of Utility Regret Bound

Definition 7 (Adam update mapping). For any $g \in \mathbb{R}^d$, define

$$h(g) = \alpha g(\mathcal{D}_{\text{val}})^\top \frac{\frac{\beta_1 \mathbf{m}_{t-1} + (1-\beta_1)g}{1-\beta_1^t}}{\sqrt{\frac{\beta_2 \mathbf{v}_{t-1} + (1-\beta_2)g \odot g}{1-\beta_2^t}} + \epsilon}.$$

Here $\mathbf{m}_{t-1}$ and $\mathbf{v}_{t-1}$ are the Adam first- and second-moment estimates at step $t-1$.

Lemma 3 (Lipschitz continuity of the Adam update mapping). Under Assumption 2, the mapping h is L_2 -Lipschitz:

$$|h(g_1) - h(g_2)| \leq L_2 \|g_1 - g_2\|_2,$$

where $L_2 = \text{poly}(\epsilon^{-1}, \alpha, G, d, \frac{1-\beta_1^2}{1-\beta_2})$.

Proof. For coordinate i , define the bias-corrected Adam moments

$$\hat{m}_{t,i} = \frac{\beta_1 m_{t-1,i} + (1-\beta_1)g_i}{1-\beta_1^t}, \quad \hat{v}_{t,i} = \frac{\beta_2 v_{t-1,i} + (1-\beta_2)g_i^2}{1-\beta_2^t}.$$

Then

$$h(g) = \alpha \sum_{i=1}^d g_i(\mathcal{D}_{\text{val}}) \frac{\hat{m}_{t,i}}{\sqrt{\hat{v}_{t,i}} + \epsilon}.$$

Differentiating with respect to g_i gives

$$\frac{\partial h}{\partial g_i} = \alpha g_i(\mathcal{D}_{\text{val}}) \frac{\hat{m}'_{t,i}(\sqrt{\hat{v}_{t,i}} + \epsilon) - \frac{\hat{m}_{t,i} \hat{v}'_{t,i}}{2\sqrt{\hat{v}_{t,i}}}}{(\sqrt{\hat{v}_{t,i}} + \epsilon)^2},$$

where $\hat{m}'_{t,i} = \frac{1-\beta_1}{1-\beta_1^t}$ and $\hat{v}'_{t,i} = 2g_i \frac{1-\beta_2}{1-\beta_2^t}$. Using $|g_i| \leq G$ and $\frac{|\hat{m}_{t,i}|}{\sqrt{\hat{v}_{t,i}}} \leq \sqrt{\frac{1-\beta_1}{1-\beta_2}}$ (Lemma 1), we obtain

$$\left| \frac{\partial h}{\partial g_i} \right| \leq \text{poly}\left(\epsilon^{-1}, \alpha, G, \frac{1-\beta_1}{1-\beta_2}\right).$$

Thus $\|\nabla h(g)\|_2 \leq \sqrt{d} \max_i \left| \frac{\partial h}{\partial g_i} \right|$, which proves the stated Lipschitz bound. $\square$

Lemma 4 (Bound on approximate utility regret). Let N be the number of source domains and T the number of acquisition rounds, each with batch size B . For domain i , let $\mathcal{D}_i \sim P_i^{BT}$ and define

$$\tilde{\mathcal{U}}_i := -\mathbb{E}_{\mathcal{D}_i} [g(\mathcal{D}_{\text{val}})^\top \Delta \theta(\mathcal{D}_i)] = \mathbb{E}_{\mathcal{D}_i} [h(g(\mathcal{D}_i))].$$

Let $\hat{i} \in \arg \max_{i \in [N]} \tilde{\mathcal{U}}_i$. Then

$$|\tilde{\mathcal{U}}_{\hat{i}} - \mathbb{E}_\pi [\tilde{\mathcal{U}}(\pi)]| \leq \mathcal{O}\left(\sqrt{\frac{N \log T}{T}}\right).$$

Proof. For a trajectory $\pi = (i_1, \dots, i_T)$, let $\mathcal{D}_{(\pi),m}$ denote a dataset of size m sampled along π . Define

$$u_i^{(m)} := \mathbb{E}[h(g(\mathcal{D}_{i,m}))], \quad u_\pi^{(m)} := \mathbb{E}[h(g(\mathcal{D}_{(\pi),m}))].$$

Then

$$\begin{aligned} |\tilde{\mathcal{U}}_{\hat{i}} - \mathbb{E}_\pi [\tilde{\mathcal{U}}(\pi)]| &= |u_{\hat{i}}^{(BT)} - \mathbb{E}_\pi [u_\pi^{(BT)}]| \\ &\leq \underbrace{\mathbb{E}_\pi [|u_{\hat{i}}^{(B)} - u_\pi^{(B)}|]}_{(1)} + \underbrace{\mathbb{E}_\pi [|u_{\hat{i}}^{(BT)} - u_{\hat{i}}^{(B)}|]}_{(2)} \\ &\quad + \underbrace{\mathbb{E}_\pi [|u_\pi^{(BT)} - u_\pi^{(B)}|]}_{(3)}. \end{aligned}$$

Term (1) is the standard N -armed bandit regret. By the UCB1 regret bound [29], we have (1) $\leq \mathcal{O}\left(\sqrt{\frac{N \log T}{T}}\right)$.

For (2) and (3), let $\bar{g}_m = m^{-1} \sum_{j=1}^m g(z_j)$. By Lemma 3,

$$\begin{aligned} & |\mathbb{E}[h(\bar{g}_{BT})] - \mathbb{E}[h(\bar{g}_B)]| \\ & \leq L_2 \mathbb{E}[\|\bar{g}_{BT} - \bar{g}_B\|_2] \leq L_2 \sqrt{\left(\frac{1}{BT} + \frac{1}{B}\right) \text{Tr}(\Sigma)}. \end{aligned}$$

Since $|g_i| \leq G$, we have $\text{Tr}(\Sigma) \leq dG^2$. Thus

$$(2), (3) \leq L_2 G \sqrt{d \left(\frac{1}{BT} + \frac{1}{B}\right)}.$$

Taking $B = T/\log T$ gives $(2), (3) = \mathcal{O}(\sqrt{\log T/T})$. Combining the three terms proves the lemma. $\square$

Proof of Theorem 2. We decompose the total expected regret $\mathcal{U}_{i_{\text{opt}}} - \mathbb{E}_\pi[\mathcal{U}(\pi)]$ into four terms:

$$\begin{aligned} \mathcal{U}_{i_{\text{opt}}} - \mathbb{E}_\pi[\mathcal{U}(\pi)] &= \underbrace{(\tilde{\mathcal{U}}_i - \mathbb{E}_\pi[\tilde{\mathcal{U}}(\pi)])}_{(i)} + \underbrace{(\mathcal{U}_{i_{\text{opt}}} - \tilde{\mathcal{U}}_{i_{\text{opt}}})}_{(ii)} \\ &\quad + \underbrace{\mathbb{E}_\pi[\tilde{\mathcal{U}}(\pi) - \mathcal{U}(\pi)]}_{(iii)} + \underbrace{(\tilde{\mathcal{U}}_{i_{\text{opt}}} - \tilde{\mathcal{U}}_i)}_{(iv)}, \end{aligned}$$

where $\hat{i} \in \arg \max_i \tilde{\mathcal{U}}_i$, which immediately yields $(iv) \leq 0$. By Lemma 4, the standard bandit regret on the approximate utility is bounded as $(i) \leq \mathcal{O}(\sqrt{N \log T/T})$.

To bound Terms (ii) and (iii), we must control the approximation error $|\tilde{\mathcal{U}}(\pi) - \mathcal{U}(\pi)|$. Because the trajectory π is adaptively chosen based on $\mathcal{D}_{\text{val}}$, we must establish a uniform bound over all N^T possible deterministic trajectories. Applying Hoeffding's inequality with a union bound, with probability at least $1 - \delta$, $\forall \pi$:

$$|\tilde{\mathcal{U}}(\pi) - \mathcal{U}(\pi)| \leq (\alpha C_1 + \alpha^2 C_2) \sqrt{\frac{T \log N + \log(4/\delta)}{2n_{\text{val}}}} + \frac{1}{2} \alpha^2 C_2.$$

This uniform bound simultaneously controls the error for the fixed domain i_{opt} (Term ii) and the adaptively selected trajectory π (Term iii). To ensure these approximation errors do not dominate the $\mathcal{O}(\sqrt{N \log T/T})$ regret, we require both the truncation error ($\frac{1}{2} \alpha^2 C_2$) and the uniform generalization error (the term scaling with $1/\sqrt{n_{\text{val}}}$) to be bounded by $\mathcal{O}(\sqrt{N \log T/T})$. Solving these two inequalities for α directly yields the condition stated in the theorem. Under this condition on α , we have $(ii) + (iii) \leq \mathcal{O}(\sqrt{N \log T/T})$. Summing all bounds completes the proof. $\square$

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